

1. Outline

This report presents a structured analysis of student social media usage and its impact on mental health, sleep, and academic performance. The report covers the following sections:

- Introduction
 - Story of Data
 - Data Splitting and Preprocessing
 - Pre-Analysis
 - In-Analysis
 - Post-Analysis and Insights
 - Data Visualizations & Charts
 - Recommendations and Observations
 - Conclusion
 - References & Appendices
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2. Introduction

Objective of the Project

The primary objective of this project is to analyze the relationship between social media usage and student wellbeing, focusing on addiction levels, academic performance, sleep patterns, and mental health outcomes.

Problem Being Addressed

With the increasing integration of social media into students' daily lives, there is growing concern about its potential negative impact on academic performance, sleep quality, and mental health. This analysis aims to identify patterns and relationships that can inform interventions.

Key Datasets and Methodologies

The dataset consists of student demographic and behavioral variables. Analysis was conducted using Microsoft Excel tools such as Pivot Tables, Charts, Conditional Formatting, and Advanced Formulas (e.g., VLOOKUP, IF, AVERAGEIF).

3. Story of Data

Data Source

The dataset represents student responses collected from academic environments, likely through surveys or institutional data collection systems.

Data Collection Process

Data was gathered through structured questionnaires capturing demographic details, social media usage habits, and wellbeing indicators.

Data Structure

- Rows represent individual student records
- Columns represent variables such as age, gender, academic level, usage hours, and mental health indicators

Important Features and Their Significance

- Avg_Daily_Usage_Hours: Measures engagement level with social media
- Most_Used_Platform: Indicates platform preference
- Sleep_Hours_Per_Night: Reflects rest patterns
- Mental_Health_Score: Indicates psychological wellbeing
- Addicted_Score: Measures level of dependency on social media
- Affects_Academic_Performance: Shows academic impact

Data Limitations or Biases

- Self-reported data may introduce bias
 - Limited geographic representation
 - Possible missing or inconsistent entries
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4. Data Splitting and Preprocessing

Data Cleaning

- Removed duplicate entries
- Standardized categorical variables
- Corrected inconsistencies in naming conventions

Handling Missing Values

- Missing values were handled using mean substitution or row exclusion where necessary

Data Transformations

- Created derived variables such as categorized addiction levels
- Grouped age ranges and usage levels

Data Splitting

- Independent Variables: Age, Gender, Academic_Level, Country, Avg_Daily_Usage_Hours, Most_Used_Platform, Relationship_Status

- Dependent Variables: Sleep_Hours_Per_Night, Mental_Health_Score, Conflicts_Over, Addicted_Score, Affects_Academic_Performance

Industry Context

This analysis falls within the academic and digital wellness sector.

Stakeholders

- Lecturers
- Students
- University Management (Chancellor)
- Government and policymakers

Value to the Industry

The analysis provides insights that can help improve student wellbeing, academic outcomes, and digital usage policies.

5. Pre-Analysis

Key Trends

- High usage of visual platforms such as Instagram and TikTok
- Average student age is approximately 21 years

Potential Correlations

- Increased social media usage may correlate with higher addiction scores
- Sleep duration may influence mental health outcomes

Initial Insights

- Undergraduate students appear to dominate usage patterns
 - Social media addiction may negatively impact academic performance
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6. In-Analysis

Unconfirmed Insights

- Higher usage hours appear linked to increased addiction scores
- Certain platforms may be associated with poorer academic outcomes

Recommendations (Preliminary)

- Monitor high-usage students
- Introduce digital awareness programs

Analysis Techniques Used in Excel

- Pivot Tables for aggregation
 - Charts for visualization
 - Conditional Formatting for pattern detection
 - Lookup functions (VLOOKUP, INDEX-MATCH)
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7. Post-Analysis and Insights

Key Findings

- Instagram is the most used platform and has the highest daily usage
- Short-form platforms dominate student engagement
- Students with higher addiction scores tend to perform worse academically
- Male and female students show similar mental health scores
- Single students experience more social media-related conflicts

Comparison with Initial Findings

Most initial assumptions were validated, particularly the relationship between high usage and addiction. However, gender differences in mental health were minimal, which was unexpected.

8. Data Visualizations & Charts

Charts and Graphs

- Bar chart: Platform usage distribution
- Line chart: Usage hours vs addiction scores
- Pie chart: Academic level distribution
- Scatter plot: Sleep vs mental health

Dashboard

An Excel dashboard was created to summarize key metrics including average usage, addiction levels, and academic performance impact.

Explanation of Visualizations

- Platform charts show dominance of Instagram and TikTok
- Scatter plots indicate a relationship between sleep and mental health
- Bar charts highlight differences across academic levels

9. Recommendations and Observations

Actionable Insights

- Introduce digital wellness education programs
- Encourage reduced screen time on high-engagement platforms
- Promote mental health awareness initiatives

Optimizations or Business Decisions

- Universities should integrate digital discipline into academic support services
- Implement sleep hygiene campaigns

Unexpected Outcomes

- Minimal gender difference in mental health scores
 - High number of non-addicted users despite heavy usage patterns
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10. Conclusion

Key Learnings

Social media significantly influences student wellbeing, particularly in areas of addiction, sleep, and academic performance.

Limitations

- Limited dataset scope
- Potential bias in self-reported data

Future Research

- Explore longitudinal data
 - Investigate intervention effectiveness
 - Include more diverse geographic samples
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11. References & Appendices

References

- Dataset source (survey data)
- Microsoft Excel documentation

Appendices

- Additional charts and tables

- Excel formulas used
- Raw dataset samples